NVIDIA cuOpt

GPU-Accelerated Decision Optimization

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The Core Challenge: Decision Bottlenecks

- **High Complexity:** Real-world logistics, supply chains, and portfolio management contain millions of variables and constraints.
- **Latency Barriers:** Traditional CPU solvers take hours to calculate optimal solutions, making real-time, dynamic decision-making impossible.
- **Operational Costs:** High solve times directly translate to elevated fuel waste, idle vehicle fleets, and inefficient resource allocation.

What is NVIDIA cuOpt?

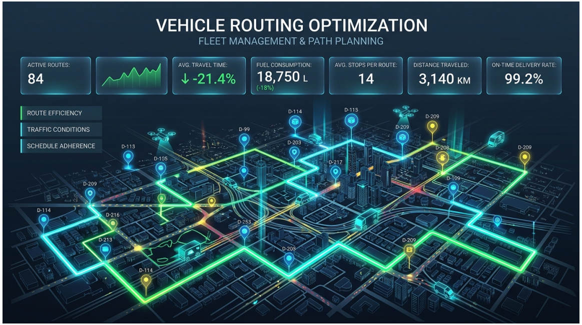
NVIDIA® cuOpt™ is an open-source, GPU-accelerated optimization engine.

Supported Mathematical Models:

- Linear Programming (LP/PDLP)
- Mixed-Integer Programming (MIP) *[Beta]*
- Quadratic Programming (QP)

Routing Specialization:

- Traveling Salesperson (TSP)
- Vehicle Routing Problems (VRP)

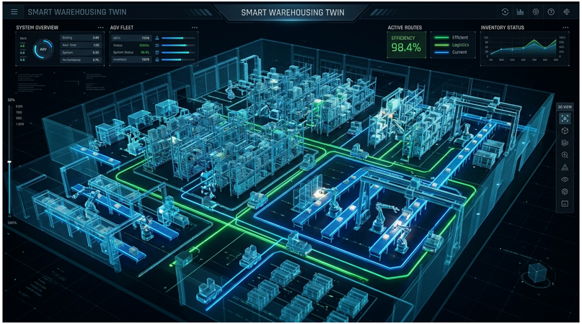


Hybrid GPU/CPU Primal Heuristics

- **GPU Offloading:** Offloads math-heavy **primal heuristics** (Feasibility Pump/Jump, Fix-and-Propagate) to thousands of parallel GPU cores.
- **CPU Tree Search:** Integrates with traditional CPU methods to perform dual-bound computations and branch-and-bound pruning.
- **Performance Gain:** Achieves up to **5,000x speedups** on large LP benchmarks and resolves objective gaps on MIPs at world-record speed.

Real-World Use Cases: Logistics & Routing

- **Last-Mile Delivery:** Optimizes routing paths to reduce fuel consumption and transit mileage (e.g., Azure Maps Integration).
- **Fleet Management:** Coordinates long-haul inbound/outbound transport, resting constraints, and schedules.
- **Omniverse Digital Twins:** Simulates physical factory warehouses in virtual environments for optimized intralogistics (e.g., BMW, SyncTwin).



Real-World Use Cases: Scheduling & Finance

- **Job Scheduling:** Dynamically assigns tasks to workers, machinery, or computing networks to optimize factory throughput. Runs near real-time re-optimizations when machines breakdown.
- **Portfolio Optimization:** Accelerates large-scale Quadratic Programming (QP) to rebalance financial assets instantly based on high-frequency market updates.

Next-Gen AI Agent Integration

- **cuOpt Agent Skills:** Open-source, modular skills compatible with common agent frameworks.
- **Natural Language Interfaces:** Translates plain English business constraints directly into optimization models.
- **Skill Cards:** Machine-readable cards providing clear documentation, security parameters, and performance validation.

Getting Started & Scale Tiers

Access Tier	Limits	Integrations	Target Use Case
API Catalog	Up to 1,000 nodes	REST API	Free prototyping
Google Colab	Free GPU Sandbox	Python API	Quick experimentation
Local Dev	Custom single-GPU	Pip/Conda/Docker	Developer builds
Enterprise	15,000+ nodes	AI Enterprise	Mission-critical production

- Seamless modeling integrations with **AMPL, CVXPY, PuLP, JuMP, and GAMSPy**.

Thank You!

Questions & Answers

nvidia.com/cuopt